

AlgoHub

SOFTWARE HOUSE

AI Agents & Automation Internship

Beginner to Advanced Agentic AI Roadmap

8 Projects | LangChain + CrewAI + AutoGen | Certificate

8 Weeks | Real-World Projects | Full Deployment

AlgoHub Software House | Official Internship Handbook | Confidential

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1. Program Overview

Welcome to the AlgoHub AI Agents & Automation Internship Program. Agentic AI is the fastest-growing area in the industry. Interns learn to design, build, and deploy autonomous AI agents that can plan, use tools, collaborate, and complete complex multi-step tasks — from a simple ReAct loop to a full multi-agent business system.

Duration	Projects	Progression	Outcome
8 Weeks	8 Complete Apps	Beginner → Advanced	Industry Certificate

This internship provides hands-on experience in:

- ◆ Agent Design

◆ LangChain

◆ Browser Automation
- ◆ Tool Use

◆ CrewAI

◆ FastAPI
- ◆ ReAct Pattern

◆ AutoGen

◆ Streamlit

2. Internship Rules & Submission Guidelines

Every project must include the following deliverables:

1	Complete Source Code	• Proper folder structure • Clean coding standards • Modular code • Proper comments
2	GitHub Repository	• README.md • Installation Guide • Dataset Information • Results & Screenshots • requirements.txt
3	ZIP File Submission	• AlgoHub_Task1_ProjectName.zip • AlgoHub_Task2_ProjectName.zip • Upload ZIP to assigned GitHub repository
4	Demo Video (5–10 min)	• Project Overview & Dataset Explanation • Data Preprocessing & Model Training • Evaluation Metrics & Live Demonstration
5	LinkedIn Submission	• Publish project post on LinkedIn • Share GitHub link, Demo Video, Screenshots • MANDATORY: Tag AlgoHub Software House
6	Deployment Requirement	• Every project deployed appropriately • FastAPI backend where applicable • Deliverables: Live URL + GitHub + ZIP + Video

3. Dataset Policy & Common Tech Stack

Dataset Policy: Do NOT request datasets. Every intern must independently search, evaluate, clean, and prepare datasets. This is mandatory to develop research and problem-solving skills.

Common Technology Stack

Programming	Python
Core Libraries	Pandas NumPy Scikit-Learn Transformers PyTorch TensorFlow
Visualization	Matplotlib Seaborn Plotly
Deployment	Streamlit FastAPI Docker
Version Control	Git GitHub
Agent Frameworks	LangChain Agents CrewAI AutoGen LlamaIndex
Tool Libraries	Tavily Search SerpAPI Playwright Selenium
LLM APIs	OpenAI GPT-4o Anthropic Claude Groq Ollama
Infra	Redis Celery Docker GitHub Actions

4. 8-Week AI Agents & Automation Roadmap

Week 1

Intro to AI Agents & Tool Use

Beginner

Concepts	Models / Tools	Skills
• What is an Agent?	• OpenAI Function Calling	• Agent Concepts
• Function Calling	• LangChain Tools	• Tool Design
• Tool Registration	• Streamlit	• LLM Integration
• Agent Loop Basics		

Objective: Build your first agent that uses tools (calculator, search, weather) via function calling.

Deploy: Streamlit App

Week 2

ReAct Agent with Web Search

Beginner

Concepts	Models / Tools	Skills
• ReAct Pattern	• LangChain ReAct	• ReAct Design
• Thought/Action/Observation	• Tavily / SerpAPI	• Web Search
• Search Tool	• GPT-4o	• Grounded Answers
• Answer Grounding	• Streamlit	

Objective: Build a ReAct (Reason + Act) agent that searches the web and answers user queries with citations.

Deploy: Streamlit Chatbot

Week 3

Multi-Tool Agent

Beginner

Concepts	Models / Tools	Skills
• Tool Chaining	• LangChain	• Multi-Tool Design
• Tool Selection Logic	• Python REPL Tool	• Safe Execution
• Error Handling	• GPT-4o	• Agent Reliability
• Sandboxed Code Execution	• FastAPI	

Objective: Build an agent with multiple tools: file reader, code executor, web search, and calculator.

Deploy: Streamlit + FastAPI

Week 4

Automated Data Pipeline Agent

Intermediate

Concepts	Models / Tools	Skills
• Agent Planning	• LangChain	• Data Automation
• CSV / API Data Ingestion	• Pandas	• Agent Planning
• Automated EDA	• Matplotlib	• Auto Reporting
• Report Generation	• GPT-4o	
	• FastAPI	

Objective: Build an agent that autonomously fetches, cleans, analyses, and reports on a dataset.

Deploy: [Streamlit Dashboard](#)

Week 5

Multi-Agent Collaboration (CrewAI)

Intermediate

Concepts	Models / Tools	Skills
• Multi-Agent Architecture	• CrewAI	• Multi-Agent Design
• Role Assignment	• GPT-4o	• Role Engineering
• Task Delegation	• Tavily Search	• Crew Orchestration
• Agent Memory	• Streamlit	
• Output Passing		

Objective: Design a crew of specialised agents (Researcher, Writer, Editor) that collaborate to produce a report.

Deploy: [Streamlit + CrewAI](#)

Week 6

Browser Automation & Web Scraping Agent

Intermediate

Concepts	Models / Tools	Skills
• Browser Control	• Playwright	• Browser Automation
• DOM Interaction	• Selenium	• Web Scraping
• Dynamic Page Handling	• BeautifulSoup	• Structured Extraction
• Anti-Bot Strategies	• GPT-4o Vision	
• Data Extraction	• FastAPI	

Objective: Build an agent that autonomously navigates websites, fills forms, and extracts structured data.

Deploy: [FastAPI + Scheduler](#)

Week 7

Self-Healing Code Agent

Int-Advanced

Concepts	Models / Tools	Skills
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• Code Generation	• GPT-4o	• Code Agent
• Execution Sandboxing	• Python subprocess	• Self-Healing AI
• Error Parsing	• pytest	• Test Automation
• Self-Correction Loop	• LangChain	
• Test Writing	• Streamlit	

Objective: Build a coding agent that writes code, runs it, catches errors, and fixes itself automatically.

Deploy: Streamlit Dev Tool

Week 8

Capstone – Autonomous Business Assistant

Advanced

Concepts	Models / Tools	Skills
• Agent Orchestration	• CrewAI / AutoGen	• Full Agentic System
• Memory & Context	• FastAPI	• Production Deployment
• Human-in-the-Loop	• Redis + Celery	• Business AI
• Task Queuing	• Streamlit	
• Production Hardening	• Docker	

Objective: Build a full autonomous assistant that manages emails, schedules tasks, generates reports, and answers business queries using a crew of specialised agents.

Deploy: Streamlit + FastAPI + Docker

5. Weekly Evaluation Criteria

Category	Weightage
Code Quality	20%
Implementation Quality	15%
Project Performance	25%
Documentation	10%
GitHub Management	10%
Deployment	10%
Demo Video & LinkedIn Submission	10%
Total	100%

6. Rewards & Certification

■ Official Internship Certificate awarded upon successful completion of all 8 projects.

- ★ Recognition Awards
- ★ LinkedIn Feature by AlgoHub
- ★ GitHub Showcase
- ★ Recommendation Opportunities
- ★ Future Hiring Consideration

7. Final Note

This roadmap is designed to gradually transform interns from beginners into industry-ready professionals. Every project replicates real production environments encountered at top software companies.

8 Complete Projects	8 GitHub Repos	8 LinkedIn Posts	8+ Live Deployments
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